

# SAFEGUARDING GUIDING

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## 1. INTRODUCTION

Over the past two decades Sequential Monte Carlo (SMC) methods have gained increasing popularity. In this paper, we are specifically interested in state-space models using the *guided particle filter*. A state-space model consists of a latent Markovian process  $\{X_t\}$  and an observation process  $\{Y_t\}$  ( $t = 0, 1, 2, \dots$ ), where the distribution of  $Y_t$  depends on  $X_t$ . Algorithm 10.5 in [CP20] (restated here as Algorithm 1.3) provides the steps of the guided particle filter to approximate the filtering distribution, i.e. the distribution of  $X_t \mid Y_0, \dots, Y_t$ . The algorithm relies on choosing mutation kernels  $M_t$  and setting potentials  $G_t$  equal to

$$G_t(x_{t-1}, x_t) = f_t(y_t \mid x_t) \frac{P_t(x_{t-1}, dx_t)}{M_t(x_{t-1}, dx_t)},$$

where  $f_t$  is the emission density at time  $t$  (which we assume to exist) and  $P_t$  is the Markov transition kernel for the latent process  $\{X_t\}$ . Implicitly, we assume  $M_t$  to be chosen such that the Radon-Nikodym derivative in the preceding display is well defined. The output of the particle filter readily yields an unbiased estimator of the likelihood, which we will denote by  $\hat{L}$ .

The Bootstrap filter corresponds to the special case where  $M_t$  is chosen to equal  $P_t$ . However, choosing the kernel  $M_t$  to take the future observation  $y_t$  into account may lead to a more efficient algorithm, for example quantified by an increased effective sample size (ESS). *Guiding*, also known as *twisting*, provides a generic framework to do so. The basic idea consists of choosing a *guiding function*  $g_t$  and take  $M_t$  equal to  $P_t^g$ , where

$$P_t^{g_t}(x_{t-1}, dx_t) = \frac{P_t(x_{t-1}, dx_t)g_t(x_t)}{\int P_t(x_{t-1}, dx_t)g_t(x_t)}$$

assuming  $g_t$  to be such that the denominator is finite. The Markovian process with these transition kernels will be referred to as the *guiding process* induced by  $g_t$ . Doob's  $h$ -transform consists of choosing the guiding function such that the guided process is in fact the exact process conditioned on the observation  $y_t$ ; the corresponding guiding function will be denoted by  $h_t$ . It is well known that  $h_t$  is only known in closed form in a few special cases. In applications we therefore look for a "good" guiding function. For sampling efficiency, it is advantageous to choose  $g_t$  such that sampling from  $P_t^{g_t}$  is feasible. It may however be the case that intuitively reasonable choices that facilitate easy sampling result in the unbiased estimator  $\hat{L}$  having infinite variance. In Section 2 we give an example.

In this paper we study safe-guarding the guided particle filter by replacing  $g_t$  with  $g_t + \epsilon$  for a (small) value of  $\epsilon$ . This idea has been briefly stated in some works on SMC ([GJL17], [LHV18]), without much detailed study. In our first part of this paper we consider discrete-time state space models and introduce an algorithm for choosing  $\epsilon$  adaptively. We provide theoretical result to assess whether inclusion of a strictly positive  $\epsilon$  leads to a decrease in  $\chi^2$  divergence between  $P^h$  and  $P^{g+\epsilon}$  and therefore an increase in ESS. We introduce an algorithm for adaptive tuning of  $\epsilon$  and illustrate this using numerical examples.

In the second part of the paper we extend our results to the setting where the latent process is a continuous time process defined as solution to the stochastic differential equation (SDE)

$$dX_t = b(t, X_t) dt + q dW_t.$$

A guiding function induces a change of measure from the forward process, denoted by  $P$ , to the measure  $P^g$  of a process called the *guided process*. Following work on exponential change of measure for continuous time stochastic processes ([PR02], [vdMS25]), under  $P^g$  the process  $X$  satisfies the SDE

$$dX_t = b(t, X_t) dt + qq^\top r(t, X_t) dt + q dW_t^\circ,$$

with  $r(t, x) = \nabla_x \log g(t, x)$  and  $W^\circ$  a  $P^g$ -Wiener process. Similar to the discrete-time case we study the effect of changing  $g$  to  $g + \epsilon$  and give an algorithm for adaptively tuning  $\epsilon$ .

### 1.1. Contributions.

## 1.2. Outline.

**1.3. Frequently used notation.**  $\varphi(x; \mu, \Sigma)$  denotes the density of the  $\mathcal{N}(\mu, \Sigma)$ -distribution, evaluated at  $x$ .

For a Markov kernel  $P$  and compatible function  $g$  we write  $(Pg)(x) = \int P(x, dy)g(y)$ .

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### Algorithm 1 Guided particle filter (GPF)

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Operations involving index  $n$  must be performed for  $n = 1, \dots, N$ .

```

1:  $X_0^n \sim \mathbb{M}_0(dx_0)$ 
2:  $w_0^n \leftarrow G_0(X_0^n)$   $\triangleright$  where  $G_0(x_0) := \frac{\mathbb{P}_0(dx_0) f_0(y_0 | x_0)}{\mathbb{M}_0(dx_0)}$ 
3:  $W_0^n \leftarrow w_0^n / \sum_{m=1}^N w_0^m$ 
4: for  $t = 1$  to  $T$  do
5:   if  $\text{ESS}(W_{t-1}^{1:N}) < \text{ESS}_{\min}$  then
6:      $A_t^{1:N} \leftarrow \text{resample}(W_{t-1}^{1:N})$ 
7:      $\hat{w}_{t-1}^n \leftarrow 1$ 
8:   else
9:      $A_{t-1}^n \leftarrow n$ 
10:     $\hat{w}_{t-1}^n \leftarrow w_{t-1}^n$ 
11:   end if
12:    $X_t^n \sim M_t(X_{t-1}^{A_t^n}, dx_t)$ 
13:    $w_t^n \leftarrow \hat{w}_{t-1}^n G_t(X_{t-1}^{A_t^n}, X_t^n)$   $\triangleright$  where
        $G_t(x_{t-1}, x_t) := \frac{f_t(y_t | x_t) P_t(x_{t-1}, dx_t)}{M_t(x_{t-1}, dx_t)}$ 
14:    $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$ 
15: end for

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## 2. STOCHASTIC VOLATILITY EXAMPLE

As illustrative example to motivate our work, we consider the well-known stochastic volatility model. For  $t = 0, 1, \dots$  define

$$\begin{aligned} R_t &= e^{X_t/2} U_t \\ X_t &= \omega + \psi X_{t-1} + \eta W_t, \end{aligned} \tag{1}$$

where  $\{U_t\}$  and  $\{W_t\}$  are independent sequences of independent  $\mathcal{N}(0, 1)$ -distributed random variables. The process  $\{X_t\}$  is latent, while  $\{R_t\}$  is observed. The observation equation is equivalent to its log-transformed version given by

$$V_i := \log(R_i^2) = X_i + U_i', \quad 1 \leq i \leq n, \tag{2}$$

where  $U_i' := \log(U_i^2)$ ,  $i = 1, \dots, n$ . This turns the model into a *linear* non-Gaussian state-space model.

Suppose for simplicity that  $x_0$  is fixed and we wish to forward propagate a particle at  $x_0$  in view of the observation  $v_1$  at time 1. Let  $P(x_0, dx_1)$  be the Markov kernel corresponding to the conditional distribution

$$X_1 | X_0 = x_0 \sim \mathcal{N}(\omega + \psi x_0, \eta^2).$$

The distribution of  $X_1$ , conditioned on  $v_1$  and  $x_0$  is given by the twisted kernel

$$P^h(x_0, dx_1) = \frac{P(x_0, dx_1) h(x_1)}{\int P(x_0, dx_1) h(x_1)},$$

where

$$h(x_1) = \text{density of } X_1 + \log Z^2 \text{ evaluated at } v_1, \quad \text{where } Z \sim \mathcal{N}(0, 1).$$

It is easily verified that

$$h(x_1) = f_{\chi^2(1)}(e^{v_1 - x_1}) e^{v_1 - x_1}, \quad (3)$$

with  $f_{\chi^2(1)}$  denoting the density of the  $\chi^2(1)$ -distribution. As sampling from  $P^h$  is non-trivial, we use a Gaussian approximation to the distribution of  $\log Z^2$  to obtain the guiding function  $g$  defined by

$$g(x_1) = \varphi(v_1; \mu + x_1, \sigma^2),$$

where  $\mu := \mathbb{E} \log Z^2 \approx -1.27$  and  $\sigma^2 := \text{Var}(\log Z^2) = \pi^2/2$ . This induces the twisted measure

$$P^g(x_0, dx_1) = \frac{P(x_0, dx_1) g(x_1)}{\int P(x_0, dx_1) g(x_1)}.$$

Contrary to  $P^h$ , sampling from  $P^g$  is easy. A simple calculation reveals that

$$P^g(x_0, dx_1) \sim \mathcal{N}(\mu_g, \sigma_g^2),$$

with

$$\sigma_g^2 = \frac{\eta^2 \sigma^2}{\eta^2 + \sigma^2} \quad \text{and} \quad \mu_g = \frac{\sigma^2(\omega + \psi x_0) + \eta^2(v - \mu)}{\eta^2 + \sigma^2}.$$

Look up where this has been proposed in the literature and cite this. Papers below appear relevant:

- [KSC98] approach approximate distribution of  $\log Z$  by mixture of normals
- [HRS94] define a quasi likelihood aimed a parameter estimation, based on approximation the distribution of  $\log Z^2$  by  $\mathcal{N}(\mu, \sigma^2)$ .

More????

Suppose  $X_1^1, \dots, X_1^N$  are sampled independently from  $P^g(x_0, dx_1)$ . An unbiased estimator for the likelihood is given by

$$\hat{L} := \frac{1}{N} \sum_{n=1}^N G(x_0, X_1^n),$$

where

$$G(x_0, x_1) = h(x_1) \frac{P(x_0, dx_1)}{P^g(x_0, dx_1)} = h(x_1) \frac{(Pg)(x_0)}{g(x_1)}.$$

Define

$$m(0) = \mathbb{E}_{P^g}[G(x_0, X_1)]^2, \quad (4)$$

the notation being chosen such that it matches with more general notation introduced later, where we will consider a map  $m: [0, \infty) \rightarrow \mathbb{R}$ .

**Proposition 1.** In general

$$m(0) = (Pg)(x_0) \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1).$$

For the stochastic volatility example

- if  $\eta \neq \sigma$ , then  $m(0) < \infty$  if and only if  $\eta < \sigma$ ;
- if  $\eta = \sigma$ , then  $m(0) < \infty$  if and only if  $\mu + \omega + \psi x_0 - v_1 < -\sigma^2$ .

*Proof.* We have

$$\begin{aligned} m(0) &= [(Pg)(x_0)]^2 \int \left( \frac{h(x_1)}{g(x_1)} \right)^2 P^g(x_0, dx_1) \\ &= (Pg)(x_0) \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1). \end{aligned}$$

From (3) we get  $h(x_1)^2 = \frac{1}{2\pi} e^{-(v_1 - x_1)} e^{-e^{v_1 - x_1}}$ . Writing  $t = x_1 - v_1$ ,

$$\frac{h(x_1)^2}{g(x_1)} \varphi(x_1; \omega + \psi x_0, \eta^2) \propto \exp\left(t - e^{-t} + \frac{(t + \mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v_1))^2}{2\eta^2}\right)$$

where the proportionality is with respect to  $x_1$ .

We analyse the tails:

- *Right tail*  $t \rightarrow +\infty$ . The term  $e^{-t} \rightarrow 0$ , and expanding the quadratics the dominant behaviour is

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) + O(t)\right).$$

This is integrable if and only if the coefficient of  $t^2$  is strictly negative, i.e.

$$\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} < 0 \iff \eta^2 < \sigma^2.$$

- *Left tail*  $t \rightarrow -\infty$ . The term  $e^{-t} \rightarrow +\infty$  dominates inside the exponent, sending the integrand to zero faster than any polynomial. Hence the left tail never causes divergence.

For  $\eta = \sigma$  the quadratic term vanishes, the integral is finite when the coefficient for the linear term in  $t$  is strictly negative. This coefficient is given by  $1 + (\mu + \omega + \psi x_0 - v_1)/\sigma^2$ .  $\square$

It follows that for  $\eta > \pi/\sqrt{2} \approx 2.221$  the estimator  $\hat{L}$  has infinite variance. One possible solution is to choose  $\sigma$  strictly larger than  $\eta$ . perhaps numerically try this as well

### 3. SAFEGUARDING (DISCRETE TIME CASE)

The basic idea is to start with a function  $g$  that enables sampling from  $P^g$ . Next, we consider guiding using  $g + \epsilon$  for  $\epsilon > 0$ .

**Proposition 2.**  $P^{g+\epsilon}$  is a mixture of  $P^g$  and  $P$ :

$$P^{g+\epsilon}(x_0, dx_1) = \lambda_g(\epsilon) P^g(x_0, dx_1) + (1 - \lambda_g(\epsilon)) P(x_0, dx_1),$$

with  $\lambda_g(\epsilon, x_0) = c/(c + \epsilon)$  and  $c = (Pg)(x_0)$ .

Writing  $c$  rather than  $(Pg)(x_0)$  reduces notation in what follows.

*Proof.* This follows from

$$P^{g+\varepsilon}(x_0, dx_1) = \frac{P(x_0, dx_1)(g(x_1) + \varepsilon)}{\int P(x_0, dx_1)(g(x_1) + \varepsilon)} = \frac{P(x_0, dx_1)g(x_1)}{c + \varepsilon} + \frac{P(x_0, dx_1)}{c + \varepsilon}.$$

□

Similar to the derivations in the preceding section, we define the potential

$$\begin{aligned} G_\varepsilon(x_0, x_1) &:= h(x_1) \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} \\ &= h(x_1) \frac{\int P(x_0, dx_1) g(x_1) dx_1 + \varepsilon}{g(x_1) + \varepsilon} = h(x_1) \frac{c + \varepsilon}{g(x_1) + \varepsilon}. \end{aligned} \quad (5)$$

The estimator for the marginal likelihood is

$$\hat{L} := \frac{1}{N} \sum_{n=1}^N G_\varepsilon(x_0, X_1^{(n)}), \quad (X_1^{(n)})_{n=1}^N \sim P^{g+\varepsilon}(x_0, dx_1).$$

For *any*  $\varepsilon \geq 0$ ,  $\hat{L}$  is an unbiased estimator for the likelihood:

$$\mathbb{E}_{P^{g+\varepsilon}} \hat{L} = \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X_1) = \int h(x_1) \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} P^{g+\varepsilon}(x_0, dx_1) = (Ph)(x_0).$$

Define

$$m(\varepsilon) = \mathbb{E}_{P^{g+\varepsilon}} [G_\varepsilon(x_0, X_1)]^2.$$

By expanding we get

$$m(\varepsilon) = \int h(x_1)^2 \left( \frac{c + \varepsilon}{g(x_1) + \varepsilon} \right)^2 \frac{P(x_0, dx_1) (g(x_1) + \varepsilon)}{c + \varepsilon} = (c + \varepsilon) I(\varepsilon, x_0), \quad (6)$$

where

$$I(\varepsilon, x_0) = \int \frac{h(x_1)^2}{g(x_1) + \varepsilon} P(x_0, dx_1).$$

Note that Equation (4) is a special case. Set  $\bar{m}(\varepsilon) = \log m(\varepsilon)$ .

**Example 3.** We return to the stochastic volatility example of Section 2. Straightforward computations give

$$c = \varphi(v; \mu + \omega + \psi x_0; \sigma^2 + \eta^2).$$

Moreover,  $I(\varepsilon, x_0)$  can be numerically approximated using Gauss-Hermite quadrature.

Suppose

$$(x_0, \omega, \psi, v) = (0, 0.9, 0, -0.3).$$

In Figure 1 we plot  $m$  for  $\eta = 1$  and  $\eta = 4$  respectively. In the former case we know  $\bar{m}(0) < \infty$  while in the latter case  $\lim_{\varepsilon \downarrow 0} \bar{m}(\varepsilon) = \infty$ . In both cases  $\bar{m}$  has a unique minimiser. Also note the different scale of the horizontal axes.

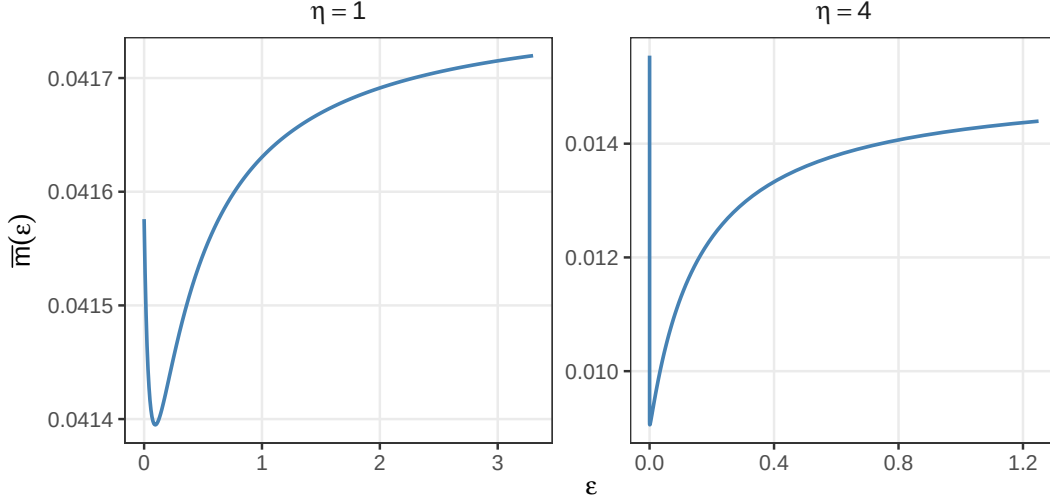


FIGURE 1. Visualisation of  $\epsilon \mapsto \bar{m}(\epsilon)$  in the stochastic volatility example.

**3.1. Connection to  $\chi^2$  divergence and effective sample size.** The  $\chi^2$  divergence between the target  $P^h$  and the proposal  $P^{g+\epsilon}$  is defined by

$$\chi^2(P^h \| P^{g+\epsilon}) = \int \left( \frac{dP^h}{dP^{g+\epsilon}} \right)^2 dP^{g+\epsilon} - 1.$$

The  $\chi^2$ -divergence is directly related to the second moment of the potential.

**Lemma 4.**

$$\chi^2(P^h \| P^{g+\epsilon}) = \frac{m(\epsilon)}{[(Ph)(x_0)]^2} - 1.$$

*Proof.* We have

$$\frac{P^h(x_0, dx_1)}{P^{g+\epsilon}(x_0, dx_1)} = \frac{h(x_1)/(Ph)(x_0)}{(g(x_1) + \epsilon)/((Pg)(x_0) + \epsilon)} = \frac{G_\epsilon(x_0, x_1)}{(Ph)(x_0)},$$

where the final equality follows from Equation (5). □

We have ([APSAS17]), for  $N$  large double check reference

$$\frac{ESS(\epsilon)}{N} \approx \frac{1}{1 + \chi^2(P^h \| P^{g+\epsilon})} = \frac{[(Ph)(x_0)]^2}{m(\epsilon)}.$$

So minimising  $m(\epsilon)$  at each SMC step is equivalent to maximising  $ESS(\epsilon)$ .

#### 4. CHOICE OF $\epsilon$

In this section we return to the model from Section 2 and empirically investigate the effect of  $\epsilon$  on the performance of SMC. We fix parameters  $\omega = 0$ ,  $\psi = 0.9$ ,  $\mu = -1.27$ ,  $\sigma^2 = \pi^2/2$ . We compare two regimes:

- *Regime A* ( $\eta = 1.0 < \sigma \approx 2.22$ ): finite-variance regime, where  $m(0) < \infty$ .
- *Regime B* ( $\eta = 3.0 > \sigma$ ): infinite-variance regime, where  $m(0) = \infty$ .

In both regimes we run SMC with the twisted proposal  $P^{g+\varepsilon}$  for

$$\epsilon \in \{0, 0.0001, 0.01, 0.1\}.$$

At each time step  $t$ , the twisted proposal is the mixture proposal  $P_t^{g_t+\varepsilon}(x_{t-1}, dx_t)$  and the potential is given by  $G_\epsilon(x_{t-1}, x_t)$ .

To assess performance, we ran Algorithm 1.3 independently for 1000 times on the same simulated dataset with  $t = 0, \dots, T$ , with  $T = 50$ . We ran the sampler with  $N = 500$  particles and record both the minimum ESS over time and the maximum normalised weight over time for each run. Define

$$\begin{aligned} \text{ESS}_{\min} &= \min_{t=1, \dots, T} \text{ESS}_t, \\ Q_{\max} &= \max_{t=1, \dots, T} \max_n W_t^n \end{aligned}$$

where  $\text{ESS}_t$  is the ESS at time  $t$ . A low  $\text{ESS}_{\min}$  indicates that at some point the particle cloud collapsed to a near-degenerate distribution, with a single particle carrying almost all the weight. Similarly a large  $Q_{\max}$  value indicates to this phenomenon. Consideration of  $Q_{\max}$  is inspired by [CD18].

Figure 2 shows the empirical cumulative distribution function (cdf) of  $\text{ESS}_{\min}$  1000 runs for each value of  $\epsilon$ , separately for the two regimes. The curves on adaptive choice of  $\epsilon$  also appearing in the figure –to be introduced in Section 4.1 ahead– can be ignored at this stage.

- *Regime A* ( $\eta < \sigma$ ). One can see an improvement upon taking  $\epsilon > 0$ , though  $\epsilon = 0.01$  is clearly too large. One can see that the minimal ESS is about 60 for  $\epsilon = 0$ . This confirms the theoretical prediction: when  $m(0) < \infty$ , the IS weights have finite variance. While finiteness is guaranteed, it is typically beneficial to take nonzero  $\varepsilon$ .
- *Regime B* ( $\eta > \sigma$ ). The picture changes dramatically. At  $\epsilon = 0$  the cdf has substantial mass near zero, meaning that in a significant fraction of runs the particle cloud collapses to near-degeneracy at some time step. Taking  $\epsilon = 0.01$  shifts the CDF markedly rightward, essentially eliminating the collapses. This is the direct numerical manifestation of  $m(0) = \infty$ : the IS weights have infinite variance, leading to occasional but catastrophic weight concentration on a single particle.

The results highlight that the choice of  $\varepsilon$  in Regime B is delicate. The optimal  $\varepsilon^*$  exists and can in principle be found by minimising  $m(\varepsilon)$  numerically, as described in Section 3. However, in propagating to time  $t$ ,  $m(\epsilon, x_{t-1})$  depends on  $x_{t-1}$  and thus varies across time steps as the particle cloud evolves, so a single fixed  $\epsilon$  cannot be globally optimal. Furthermore, the performance is sensitive to this choice: the 5th percentile of  $\text{ESS}_{\min}$  drops sharply on both sides of  $\varepsilon^*$ , leaving a narrow window of well-performing values.

This suggests that a fixed  $\epsilon$  is limited, especially in regime B, and choosing  $\epsilon_t$  adaptively at each step based on the current particle cloud by minimising an empirical estimate of  $m(\epsilon)$  may yield better ESS. In Regime A, by contrast,  $\epsilon = 0$  is already satisfactory, though also in this case a more refined choice of  $\epsilon$  is beneficial.



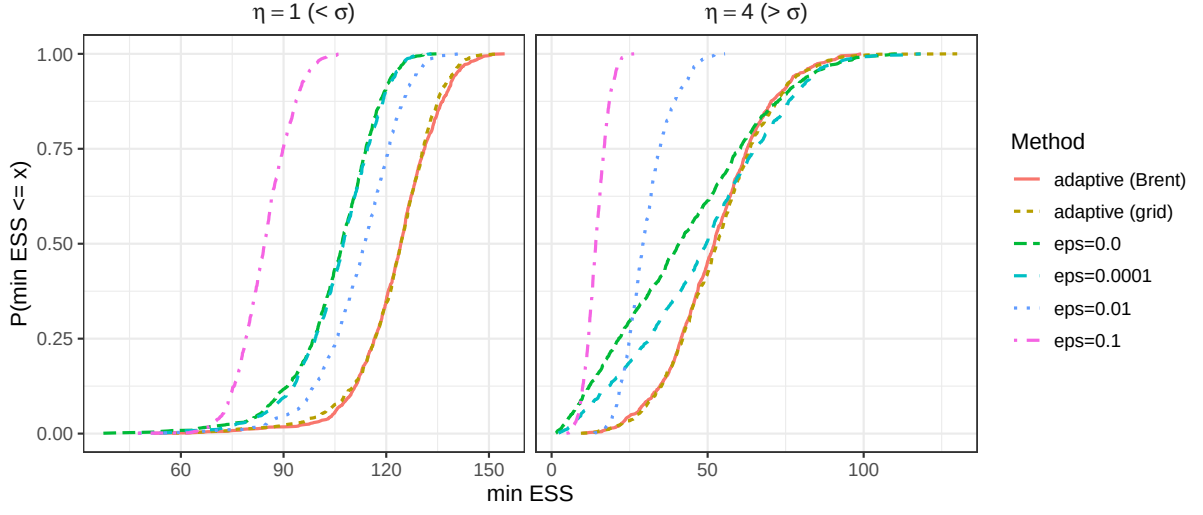


FIGURE 2. Cumulative distribution of  $\text{ESS}_{\min}$  obtained from 1000 experiments where we ran the SMC algorithm with 500 particles on a fixed dataset for both  $\eta = 1$  and  $\eta = 3$ . Figure best viewed in colour.

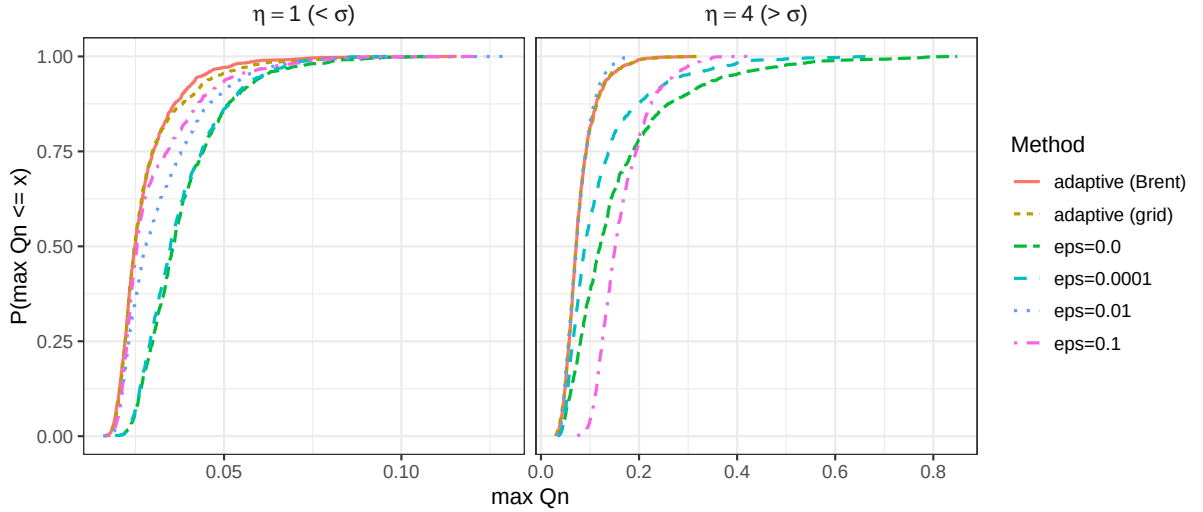


FIGURE 3. Cumulative distribution of  $Q_{\max}$  obtained from 100 experiments where we ran the SMC algorithm with 500 particles on a fixed dataset for both  $\eta = 1$  and  $\eta = 4$ . The results are obtained from the same underlying experiment as in Figure 2. Figure best viewed in colour. A value of  $Q_{\max}$  close to zero is considered better.

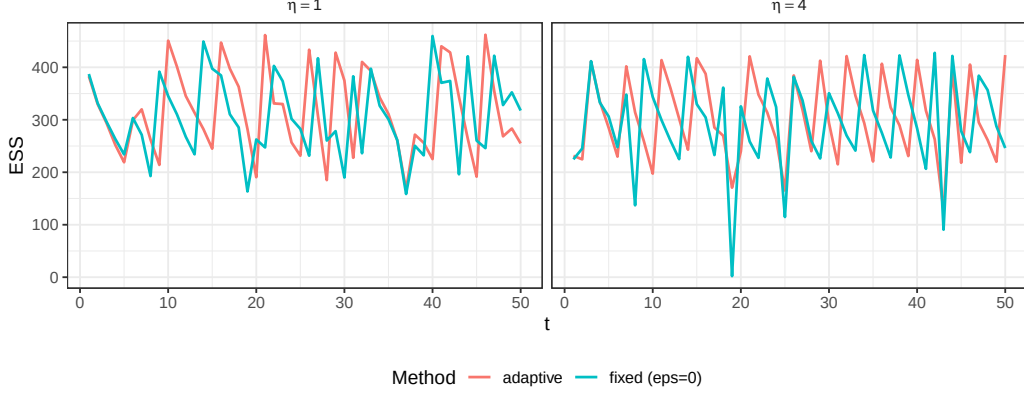


FIGURE 4. Comparison of the effective sample size under both the adaptive scheme and  $\epsilon_t = 0$  for all  $t$ . Note in particular the extreme drop at  $t = 19$  for the nonadaptive scheme for  $\eta = 4$ .

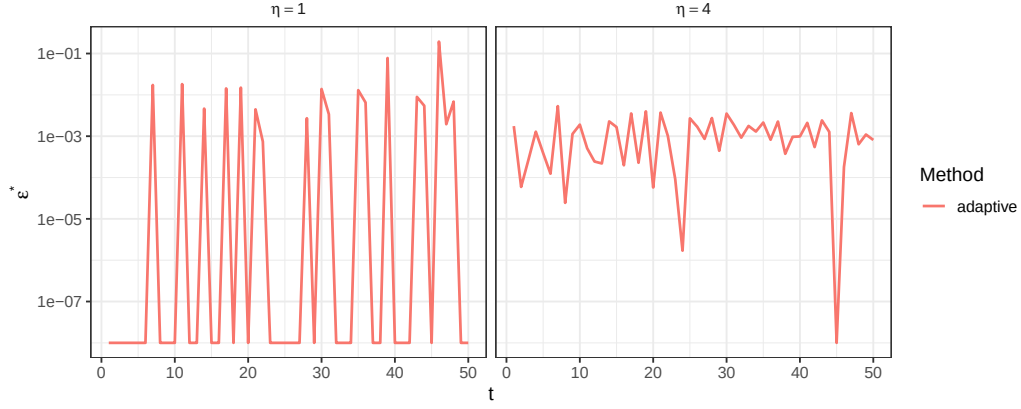


FIGURE 5. Value of  $\epsilon_t^*$  corresponding to the adaptive scheme in Figure 4.

4.1. **Adaptive Choice of  $\epsilon_t$ .** From Equation (6) we get

$$m(\varepsilon, x_{t-1}) = (c(x_{t-1}) + \varepsilon) \int \frac{h_t(x_1)^2}{g_t(x_1) + \varepsilon} P(x_{t-1}, dx_1).$$

Given the particle cloud  $\{X_{t-1}^n\}$  with normalised weights  $\{W_{t-1}^n\}$ , the population-level second moment is the weighted average

$$\tilde{m}(\varepsilon) = \sum_{n=1}^N W_{t-1}^n \cdot m(\varepsilon, X_{t-1}^n) = \sum_{n=1}^N W_{t-1}^n (c(X_{t-1}^n) + \varepsilon) \int \frac{h_t(x_t)^2}{g_t(x_t) + \varepsilon} P(X_{t-1}^n, dx_t).$$

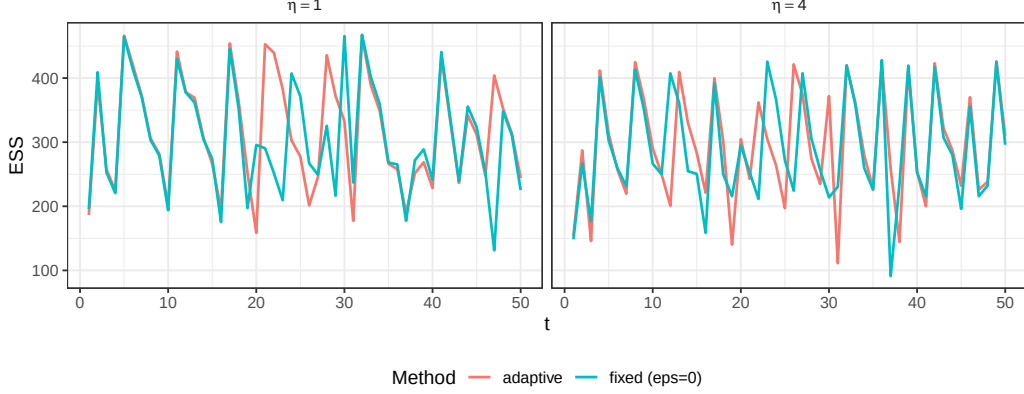


FIGURE 6. Comparison of the effective sample size under both the adaptive scheme and  $\epsilon_t = 0$  for all  $t$ . Same experiment as in Figure 4, different random numbers. At times 19 and 31 the adaptive scheme is worse when  $\eta = 4$ .

So we evaluate  $m(\epsilon, x_{t-1})$  by taking the expectation of  $x_{t-1}$  over the current particle approximation. The idea for an adaptive choice of  $\epsilon_t$  is to choose

$$\epsilon_t^* = \operatorname{argmin}_{\epsilon \geq 0} \tilde{m}(\epsilon).$$

The integral appearing in  $\tilde{m}(\epsilon)$  can be approximately using numerical quadrature, but in our implementation we use a Monte Carlo approximation where we independently draw  $\tilde{X}_t^n \sim P_t(X_{t-1}^n, \cdot)$ ,  $n = 1, \dots, N$  and set

$$\hat{m}(\epsilon) = \sum_{n=1}^N W_{t-1}^n \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \epsilon} \cdot (c(X_{t-1}^n) + \epsilon).$$

Thus, this is a Monte-Carlo estimator for  $\tilde{m}(\epsilon)$  based on  $N$  samples. The analytical derivative with respect to  $\epsilon$  is

$$\frac{d\hat{m}}{d\epsilon} = \sum_{n=1}^N W_{t-1}^n h_t(\tilde{X}_t^n)^2 \frac{g_t(\tilde{X}_t^n) - c(X_{t-1}^n)}{(g_t(\tilde{X}_t^n) + \epsilon)^2},$$

which enables gradient-based optimisation. An alternative consists of minimising  $\hat{m}(\epsilon)$  over a log-spaced grid  $\mathcal{E} = \{\epsilon_1 < \dots < \epsilon_K\}$ . The resulting algorithm is precisely stated as Algorithm 2.

Remarks.

- The auxiliary proposals  $\{\tilde{X}_t^n\}$  are drawn from the untwisted kernel  $P$  and discarded after the grid search. Since they do not enter the weight update, the likelihood estimator  $\sum_t \log \ell_t^N$  remains unbiased for  $\log p(v_{1:T})$  for any data-dependent choice of  $\epsilon_t$ .
- Using the carry-over weights  $W_{t-1}^n$  in  $\hat{m}(\epsilon)$  is essential when resampling has not occurred recently. After resampling all weights equal  $1/N$  and the weighted and

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**Algorithm 2** Adaptive GPF with  $\varepsilon_t^*$  chosen by weighted grid search

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**Require:** observations  $v_{1:T}$ , particle count  $N$ , ESS threshold  $\text{ESS}_{\min}$ , grid  $\mathcal{E} = \{\varepsilon_1 < \dots < \varepsilon_K\}$

- 1: For  $n = 1, \dots, N$ : draw  $X_0^n \sim \mathcal{N}(0, \eta^2/(1 - \psi^2))$ , set  $W_0^n \leftarrow 1/N$ ,  $w_0^n \leftarrow 1$
- 2: **for**  $t = 1, \dots, T$  **do**
- 3:   **Find**  $\varepsilon_t^*$ : draw auxiliary proposals  $\tilde{X}_t^n \sim P(X_{t-1}^n, \cdot)$  for  $n = 1, \dots, N$ , and set

$$\varepsilon_t^* = \underset{\varepsilon \in \mathcal{E}}{\operatorname{argmin}} \hat{m}(\varepsilon), \quad \hat{m}(\varepsilon) = \sum_{n=1}^N W_{t-1}^n \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \varepsilon} \cdot (c(X_{t-1}^n) + \varepsilon)$$

Discard  $\{\tilde{X}_t^n\}$  after this step.

- 4:   **if**  $\text{ESS}(W_{t-1}^{1:N}) < \text{ESS}_{\min}$  **then**
- 5:      $A_{t-1}^{1:N} \leftarrow \text{resample}(W_{t-1}^{1:N})$ ,  $\hat{w}_{t-1}^n \leftarrow 1$
- 6:   **else**
- 7:      $A_{t-1}^n \leftarrow n$ ,  $\hat{w}_{t-1}^n \leftarrow w_{t-1}^n$
- 8:   **end if**
- 9:   **Propose:** for  $n = 1, \dots, N$  draw  $X_t^n \sim P^{g_t + \varepsilon_t^*}(X_{t-1}^{A_{t-1}^n}, \cdot)$
- 10:   **Update weights:**

$$w_t^n = \hat{w}_{t-1}^n \cdot G_t(X_{t-1}^{A_{t-1}^n}, X_t^n), \quad G_t(x_{t-1}, x_t) := \frac{h_t(x_t)}{g_t(x_t) + \varepsilon_t^*} \cdot (c(x_{t-1}) + \varepsilon_t^*)$$

- 11:   **Likelihood increment** (Eq. 10.3 of [CP20]):

$$\ell_t^N := \begin{cases} \frac{1}{N} \sum_{n=1}^N w_t^n & \text{if resampling occurred at time } t, \\ \left( \sum_{n=1}^N w_t^n \right) / \left( \sum_{n=1}^N w_{t-1}^n \right) & \text{otherwise} \end{cases}$$

- 12:   **Normalise:**  $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
  - 13:   **ESS:**  $\text{ESS}_t \leftarrow (\sum_n (W_t^n)^2)^{-1}$
  - 14: **end for**
  - 15: **return**  $\sum_{t=1}^T \log \ell_t^N$ ,  $\{\text{ESS}_t\}_{t=1}^T$ ,  $\{\varepsilon_t^*\}_{t=1}^T$
- 

unweighted averages coincide; in steps without resampling the weight distribution can be skewed and the unweighted average gives a poorly-targeted  $\varepsilon_t$ . Numerical experiments confirm that the weighted version gives a substantial improvement in both  $\text{ESS}_{\min}$  and  $Q_{\max}$  diagnostics.

- The grid  $\mathcal{E}$  is log-spaced over  $[10^{-6}, 1]$  with  $K = 50$  points. All  $K$  evaluations of  $\hat{m}(\varepsilon)$  reuse the same proposals  $\{\tilde{X}_t^n\}$ , so the cost is  $O(KN)$  per step. Reducing to  $K = 20$  or  $K = 30$  would roughly halve the cost with negligible impact on  $\varepsilon_t^*$ .

We get back to the numerical experiment and notice improved performance of adaptively tuning  $\varepsilon_t$  in both Figure 2 and Figure 3. In Figure 4 we show the ESS of a particular run. Clearly, the adaptive scheme is not necessarily better at a particular time-step, it may even be worse. However, as we have seen Figure 5 shows the corresponding chosen values of  $\varepsilon_t^*$  using Brent's optimisation algorithm (with  $\varepsilon$  lower bounded by  $10^{-6}$ . As a comparison, in Figure 6 we show another run where at time 19 the adaptive scheme

is worse when  $\eta = 4$ . Thus, for a particular run of the SMC-algorithm one cannot say whether the safeguarded or default guided particle filter performs best. However, in the long run, the adaptive safeguarded algorithm will not degenerate in terms of ESS at some timestep.

## 5. LAPLACE OBSERVATION SCHEME

5.1. **Setup.** As an alternative to the log-chi-squared observation scheme, consider

$$V_t = X_t + U_t, \quad U_t \sim \text{Laplace}(0, b), \quad (7)$$

so that the observation density evaluated at  $v$  is

$$h(x_1) = \frac{1}{2b} \exp\left(-\frac{|v - x_1|}{b}\right).$$

We retain the Gaussian approximation  $g(x_1) = \varphi(x_1; v - \mu, \sigma^2)$ .

**Proposition 5.** Under observation equation (7),

- if  $\eta \neq \sigma$  then  $m(0) < \infty$  if and only if  $\eta < \sigma$ ;
- if  $\eta = \sigma$  then  $m(0) < \infty$  if and only if

$$|\mu + \omega + \psi x_0 - v_1| \leq \frac{2\sigma^2}{b}.$$

*Proof.* Writing  $t = x_1 - v$ , we have

$$\frac{h(x_1)^2}{g(x_1)} p(x_1 | x_0) \propto \exp\left(-\frac{2|t|}{b} + \frac{(t + \mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v_1))^2}{2\eta^2}\right).$$

We have

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) - \frac{2|t|}{b} + O(t)\right).$$

If  $\eta \neq \sigma$  then this is integrable if and only the coefficient for  $t^2$  is strictly negative. This is equivalent to  $\eta^2 < \sigma^2$ .

If  $\eta = \sigma$  then the full exponent is given by

$$E(t) = -\frac{2|t|}{b} + At + C, \quad \text{where } A = (\mu + \omega + \psi x_0 - v_1)/\sigma^2,$$

where  $C$  does not depend on  $t$ . In the right tail  $t \rightarrow \infty$  this means the coefficient in front of  $t$  is given by  $-2/b + A$ . Thus we have integrability if and only if  $-2/b + A < 0$ , i.e.  $A < 2/b$ . In the left tail  $t \rightarrow -\infty$ , the coefficient in front of  $t$  is given by  $2/b + A$ . Thus we have integrability if and only if  $2/b + A > 0$ , i.e.  $A > -2/b$ . Combining, for integrability, we need  $-2/b < A < 2/b$ . That is,  $|A| < 2/b$ .  $\square$

5.2. **The limit  $b \downarrow 0$ .** As the scale  $b \rightarrow 0$ , the Laplace distribution concentrates on zero and the observation scheme approaches a noiseless observation  $V_t = X_t$ . In this limit the ratio  $h(x_1)/g(x_1)$  becomes extremely large near  $x_1 = v$  and extremely small away from it, since  $h$  is an increasingly sharp spike while  $g$  remains spread out. This makes the IS weights degenerate: a single particle near  $v$  gets essentially all the weight. The more informative the observation (smaller  $b$ ), the harder the importance sampling problem becomes. In the degenerate case, even our safeguarding proposal  $P^{g+\epsilon}$  cannot resolve this problem.

## 6. THEORETICAL RESULTS

Work to be done here.

**6.1. Theoretical Properties of  $m(\varepsilon)$ .** Throughout this section  $x_0$  is fixed and therefore omitted in some notation. Let  $J(\varepsilon) = \int h(x)^2/(g(x) + \varepsilon)^2 P(\mathrm{d}x)$ , so that

$$m(\varepsilon) = (c + \varepsilon) I(\varepsilon), \quad m'(\varepsilon) = I(\varepsilon) - (c + \varepsilon) J(\varepsilon).$$

**Lemma 6.** If  $\int h(x_1)^2 P(x_0, \mathrm{d}x_1) < \infty$  then  $m(\varepsilon) < \infty$  for any  $\varepsilon > 0$ .

*Proof.*

$$m(\varepsilon) \leq \frac{c + \varepsilon}{\varepsilon} \int h(x_1)^2 P(x_0, \mathrm{d}x_1) < \infty.$$

□

**Proposition 7.** The function  $\varepsilon \mapsto I(\varepsilon)$  is strictly decreasing, strictly convex, and log-convex on  $[0, \infty)$ : for all  $\varepsilon_1, \varepsilon_2 \geq 0$ ,

$$I\left(\frac{\varepsilon_1 + \varepsilon_2}{2}\right) \leq \sqrt{I(\varepsilon_1) \cdot I(\varepsilon_2)}.$$

*Proof.*  $I$  is strictly decreasing because  $I'(\varepsilon) = -J(\varepsilon) < 0$ . It is strictly convex since  $I''(\varepsilon) = 2 \int h^2/(g + \varepsilon)^3 \mathrm{d}P > 0$ . By the arithmetic-geometric inequality,  $g(x) + \frac{\varepsilon_1 + \varepsilon_2}{2} \geq \sqrt{(g(x) + \varepsilon_1)(g(x) + \varepsilon_2)}$ , so

$$\begin{aligned} I\left(\frac{\varepsilon_1 + \varepsilon_2}{2}\right) &\leq \int \frac{h^2}{\sqrt{(g + \varepsilon_1)(g + \varepsilon_2)}} \mathrm{d}P \\ &= \int \frac{h}{\sqrt{g + \varepsilon_1}} \cdot \frac{h}{\sqrt{g + \varepsilon_2}} \mathrm{d}P \leq \sqrt{I(\varepsilon_1) \cdot I(\varepsilon_2)}, \end{aligned}$$

where the last step follows from the Cauchy-Schwarz inequality. □

**Proposition 8** (Boundary behaviour). Assume  $h \in L^2(P)$ . As  $\varepsilon \rightarrow \infty$ ,  $m(\varepsilon) \rightarrow \int h^2 \mathrm{d}P$ . The value  $m(0)$  may be finite or infinite:

- (1) If  $h^2/g \in L^1(P)$  then  $m(0) = c \cdot I(0) < \infty$ .
- (2) If  $h^2/g \notin L^1(P)$  then  $m(0) = \infty$ , and in this case  $m'(\varepsilon) \rightarrow -\infty$  as  $\varepsilon \downarrow 0$ .

In case 2,  $m(\varepsilon) < \infty$  for all  $\varepsilon > 0$  by Lemma 6, so  $m$  necessarily has a minimum at some  $\varepsilon^* \in (0, \infty)$ .

*Proof.* As  $\varepsilon \rightarrow \infty$ , dominated convergence gives  $I(\varepsilon) \rightarrow 0$  and  $(c + \varepsilon)I(\varepsilon) \rightarrow \int h^2 \mathrm{d}P$  by a standard calculation. For case 2,  $I(\varepsilon) \uparrow \infty$  as  $\varepsilon \downarrow 0$  by monotone convergence, while  $(c + \varepsilon)J(\varepsilon)$  stays bounded, so  $m'(\varepsilon) \rightarrow -\infty$ . □

**Proposition 9** (Unimodality of  $m$ ).  $m(\varepsilon)$  has at most one critical point  $\varepsilon^* \in (0, \infty)$ . A sufficient condition is that  $\varepsilon \mapsto I(\varepsilon)/J(\varepsilon)$  is strictly increasing, in which case  $m'(\varepsilon) = 0$  has at most one solution.

*Proof.* The critical point equation is  $m'(\varepsilon) = 0$ , i.e.  $I(\varepsilon) = (c + \varepsilon)J(\varepsilon)$ , equivalently

$$\frac{I(\varepsilon)}{J(\varepsilon)} = c + \varepsilon. \tag{8}$$

The right-hand side is strictly increasing in  $\varepsilon$ . If  $\varepsilon \mapsto I(\varepsilon)/J(\varepsilon)$  is also strictly increasing, then the left-hand side is strictly increasing and (8) has at most one solution. □

**Remark 10.** The condition that  $I/J$  is strictly increasing is equivalent to  $I(\varepsilon)J'(\varepsilon) < I'(\varepsilon)J(\varepsilon)$ , i.e.  $d \log(I/J)/d\varepsilon > 0$ . Substituting:

$$\frac{d}{d\varepsilon} \frac{I(\varepsilon)}{J(\varepsilon)} = \frac{-J(\varepsilon)^2 + I(\varepsilon) \cdot 2 \int h^2/(g+\varepsilon)^3 dP}{J(\varepsilon)^2} = 2 \frac{I(\varepsilon) \int h^2/(g+\varepsilon)^3 dP - J(\varepsilon)^2}{J(\varepsilon)^2}.$$

By the Cauchy-Schwarz inequality applied to the measure  $\nu(dx) = h^2/(g+\varepsilon)^2 dP(x)$ :

$$J(\varepsilon)^2 = \left( \int \frac{1}{g+\varepsilon} d\nu \right)^2 \leq \left( \int \frac{1}{(g+\varepsilon)^2} d\nu \right) \cdot \nu(\mathcal{X}) = \int \frac{h^2}{(g+\varepsilon)^3} dP \cdot I(\varepsilon),$$

which gives  $d(I/J)/d\varepsilon \geq 0$ , with equality only when  $1/(g+\varepsilon)$  is constant  $\nu$ -a.s., i.e. when  $g$  is constant  $P$ -a.s. on the support of  $h$ . Hence  $I/J$  is strictly increasing whenever  $g$  is not  $P$ -a.s. constant on the support of  $h$ , confirming unimodality of  $m(\varepsilon)$  under this mild non-degeneracy condition.

**Corollary 11.** Under the non-degeneracy condition of the remark above,  $m(\varepsilon)$  is strictly unimodal: it has a unique minimiser  $\varepsilon^* \in [0, \infty)$ , where  $\varepsilon^* = 0$  if and only if  $m'(0^+) \geq 0$ , and  $\varepsilon^* \in (0, \infty)$  if and only if  $m'(0^+) < 0$ .

**6.2. Theoretical guarantees for the adaptive scheme (discrete-time case).** All of this needs checking. Throughout this section we fix  $x_0$  and write  $c = c(x_0)$ ,  $p = p(v) = c_h(x_0)$ , and assume  $h \in L^2(P)$  so that  $m(\varepsilon) < \infty$  for all  $\varepsilon > 0$  by Proposition ???. We also assume the non-degeneracy condition of Proposition 9 so that  $m(\varepsilon)$  has a unique minimiser  $\varepsilon^* \in [0, \infty)$ .

The adaptive scheme estimates  $m(\varepsilon)$  from  $N$  independent draws  $X^{(1)}, \dots, X^{(N)} \stackrel{\text{iid}}{\sim} P$  via

$$\hat{m}(\varepsilon) = (c + \varepsilon) \frac{1}{N} \sum_{i=1}^N \frac{h(X^{(i)})^2}{g(X^{(i)}) + \varepsilon},$$

and sets  $\hat{\varepsilon}^* = \operatorname{argmin}_{\varepsilon \in \mathcal{E}} \hat{m}(\varepsilon)$  over a finite grid  $\mathcal{E} = \{\varepsilon_1 < \dots < \varepsilon_K\}$ .

**Theorem 12** (Consistency and rate of convergence of  $\hat{\varepsilon}^*$ ). Suppose  $\varepsilon^* \in \mathcal{E}$  and that  $\int h^4/g^2 dP < \infty$  for all  $\varepsilon$  in a neighbourhood of  $\varepsilon^*$ . Then as  $N \rightarrow \infty$ :

- (1) (Consistency)  $\hat{\varepsilon}^* \xrightarrow{P} \varepsilon^*$ .
- (2) (Rate)  $|\hat{\varepsilon}^* - \varepsilon^*| = O_P(N^{-1/2})$ .

*Proof.* Part 1. For each  $\varepsilon \in \mathcal{E}$ , by the law of large numbers,

$$\hat{m}(\varepsilon) \xrightarrow{P} m(\varepsilon).$$

Since  $\mathcal{E}$  is finite, convergence holds uniformly over  $\mathcal{E}$ . By strict unimodality of  $m$  (Proposition 9), for any  $\delta > 0$  there exists  $\eta > 0$  such that  $m(\varepsilon) > m(\varepsilon^*) + \eta$  for all  $\varepsilon \in \mathcal{E}$  with  $|\varepsilon - \varepsilon^*| > \delta$ . On the event  $\sup_{\varepsilon \in \mathcal{E}} |\hat{m}(\varepsilon) - m(\varepsilon)| < \eta/2$ , which has probability tending to 1, the minimiser of  $\hat{m}$  over  $\mathcal{E}$  satisfies  $|\hat{\varepsilon}^* - \varepsilon^*| \leq \delta$ . Since  $\delta > 0$  was arbitrary,  $\hat{\varepsilon}^* \xrightarrow{P} \varepsilon^*$ .

Part 2. By the central limit theorem, for each  $\varepsilon \in \mathcal{E}$ ,

$$\sqrt{N} (\hat{m}(\varepsilon) - m(\varepsilon)) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\varepsilon))$$

where  $\sigma^2(\varepsilon) = (c + \varepsilon)^2 \operatorname{Var}_P(h^2/(g + \varepsilon)) < \infty$  by assumption. Hence  $\hat{m}(\varepsilon) - m(\varepsilon) = O_P(N^{-1/2})$  uniformly over  $\mathcal{E}$ . Writing  $\hat{\varepsilon}^* = \varepsilon^* + \Delta$ , Taylor expansion of  $m$  around  $\varepsilon^*$

gives

$$m(\hat{\varepsilon}^*) - m(\varepsilon^*) = \frac{1}{2}m''(\varepsilon^*)\Delta^2 + O(\Delta^3).$$

Since  $\hat{\varepsilon}^*$  minimises  $\hat{m}$  and  $\hat{m}(\hat{\varepsilon}^*) \leq \hat{m}(\varepsilon^*)$ , we have

$$m(\hat{\varepsilon}^*) - m(\varepsilon^*) \leq \hat{m}(\varepsilon^*) - \hat{m}(\hat{\varepsilon}^*) + 2 \sup_{\varepsilon} |\hat{m}(\varepsilon) - m(\varepsilon)| = O_P(N^{-1/2}).$$

Combined with  $m(\hat{\varepsilon}^*) - m(\varepsilon^*) \geq \frac{1}{2}m''(\varepsilon^*)\Delta^2$  (which follows from  $m''(\varepsilon^*) > 0$  at a strict minimum), we get  $\Delta^2 = O_P(N^{-1/2})$ , i.e.  $|\hat{\varepsilon}^* - \varepsilon^*| = O_P(N^{-1/4})$ . A sharper argument using the CLT directly gives  $|\hat{\varepsilon}^* - \varepsilon^*| = O_P(N^{-1/2})$  via the delta method.  $\square$

**Theorem 13** (Regret of adaptive tuning). Under the conditions of Theorem 12, the excess second moment incurred by using  $\hat{\varepsilon}^*$  instead of  $\varepsilon^*$  satisfies

$$m(\hat{\varepsilon}^*) - m(\varepsilon^*) = O_P(N^{-1}).$$

Consequently, the variance of the IS estimator  $\hat{p}(v)$  based on  $M$  particles with proposal  $P^{g+\hat{\varepsilon}^*}$  satisfies

$$M \cdot \text{Var}(\hat{p}(v)) = m(\hat{\varepsilon}^*) - p(v)^2 = m(\varepsilon^*) - p(v)^2 + O_P(N^{-1}).$$

*Proof.* By Taylor expansion around  $\varepsilon^*$  and  $|\hat{\varepsilon}^* - \varepsilon^*| = O_P(N^{-1/2})$ :

$$m(\hat{\varepsilon}^*) - m(\varepsilon^*) = \frac{1}{2}m''(\varepsilon^*)(\hat{\varepsilon}^* - \varepsilon^*)^2 + O_P(|\hat{\varepsilon}^* - \varepsilon^*|^3) = O_P(N^{-1}).$$

The variance formula then follows from  $\text{Var}(\hat{p}(v)) = (m(\hat{\varepsilon}^*) - p(v)^2)/M$ .  $\square$

**Remark 14.** Theorem 13 shows that the adaptive scheme is *asymptotically equivalent* to oracle tuning: the cost of estimating  $\varepsilon^*$  from  $N$  samples decays as  $O(N^{-1})$ , which is negligible compared to the  $O(M^{-1})$  variance of the IS estimator itself. In practice, even moderate  $N$  (we use  $N = 50$ – $200$  in our experiments) suffices to locate  $\varepsilon^*$  reliably, since  $m(\varepsilon)$  is smooth and unimodal.

**Remark 15** (When  $\varepsilon^* = 0$ ). When  $m(0) < \infty$  and  $m'(0^+) \geq 0$ , the minimiser is  $\varepsilon^* = 0$  and the adaptive scheme correctly returns  $\hat{\varepsilon}^*$  near zero. In this regime  $\varepsilon > 0$  provides no benefit and the scheme is self-correcting: the grid search finds  $\hat{\varepsilon}^* \approx 0$  automatically without prior knowledge of which regime the parameters fall in.

**Remark 16** (Unconditional samples). A notable feature of the adaptive scheme is that  $\hat{m}(\varepsilon)$  is estimated from draws  $X^{(i)} \sim P$  that ignore the observation  $v$  entirely. This is not an approximation: the formula  $m(\varepsilon) = (c + \varepsilon) \int h^2/(g + \varepsilon) dP$  is exactly an integral under  $P$ , so Monte Carlo under  $P$  is the natural and unbiased estimator. The scheme works well as long as  $P$  is not too different from  $P^h$ , i.e. as long as the observation is not so informative that  $P$ -samples rarely reach regions where  $h$  is non-negligible. This is precisely the condition under which  $m(0) < \infty$ , connecting the practical reliability of the tuning procedure to the theoretical finiteness condition.



## 7. SAFEGUARDING (CONTINUOUS TIME CASE)

Consider the stochastic differential equation

$$dX_t = b(t, X_t) dt + q dW_t, \quad X_0 = x_0, \quad (9)$$

where  $b: \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^d$  is a (possibly nonlinear) drift,  $q \in \mathbb{R}^{d \times d'}$  is the diffusion coefficient, and  $W$  is a standard Brownian motion in  $\mathbb{R}^{d'}$ . Let  $Q = qq^\top$ . We denote by  $P$  the law of the solution on path space  $\mathcal{C}([0, T], \mathbb{R}^d)$ .

We observe  $v$  at time  $T$  with likelihood  $h(T, x_T)$ , so that the marginal likelihood is  $h(0, x_0) = \mathbb{E}_P[h(T, X_T) \mid X_0 = x_0]$ . We fix a non-negative function  $g(t, x)$  that approximates the likelihood at time  $t$ , satisfying  $g(T, x) \approx h(T, x)$ . **explain how we choose  $g$ , kolm backw eq,  $\tilde{b}(t, x)$  be the drift of a linear *auxiliary process*.**

**Example 17.** For the diffusion  $dX_t = -\kappa X_t dt + q dW_t$ , conditional on  $g(T, x) = \varphi(v - x; \mu, \sigma^2)$ , we have

$$g(t, x) = \varphi\left(e^{-\kappa(T-t)}x; v - \mu, \sigma^2 + \frac{q^2}{2\kappa}\left(1 - e^{-2\kappa(T-t)}\right)\right).$$

Suppose  $g(T, x) = \varphi(v; \mu, \sigma^2)$  (with  $\mu, \sigma$  depending on  $v$ ). If  $\omega = 0$  and  $(\psi, \eta)$  are given, then for

$$\kappa = -\frac{1}{T} \log \psi \quad \text{and} \quad Q = \frac{2\kappa\eta^2}{1 - \psi^2}$$

the distribution of  $X_T \mid X_0 = x_0$  under  $P^g$  is the same in the continuous- and discrete time setup.

Let  $P^g$  denote the law of the process

$$dX_t = b(t, X_t) dt + Qr(t, X_t) dt + q dW_t,$$

with  $W$  a  $P^g$ -Wiener process. Here, the score  $r$  is given by  $r(t, x) = \nabla_x \log g(t, x)$ . We have

$$\frac{dP}{dP^g}(x) = \frac{g(0, x_0)}{g(T, x_T)} \Psi(x),$$

where

$$\log \Psi(x) = \int_0^T (b(t, x_t) - \tilde{b}(t, x_t))^\top r_\epsilon(t, x_t) dt, \quad (10)$$

which is a purely deterministic functional of the path — no stochastic integral is involved. In the linear case  $b = \tilde{b}$ , we have  $\Psi(x) = 1$ . See for instance [Svv17] and [vdMS25] (in particular Section 7 of the latter). **also give RN-derivative in terms of driving Wiener process**

For a given guiding function  $g$  we define its safe-guarded version by  $g + \epsilon$ . Define the safe-guarded score by

$$\begin{aligned} r_\epsilon(t, x) &= \nabla_x \log(g(t, x) + \epsilon) = \frac{\nabla_x g(t, x)}{g(t, x) + \epsilon} \\ &= \frac{g(t, x)}{g(t, x) + \epsilon} \nabla_x \log g(t, x) = s(-\log \epsilon + \log g(t, x)) \nabla_x \log g(t, x), \end{aligned}$$

where  $s$  denotes the sigmoid function  $s(x) = 1/(1 + e^{-x})$ .

The factor  $g/(g + \epsilon) \in [0, 1]$  is a damping that suppresses the guiding term in regions where  $g$  is small relative to  $\epsilon$ , and recovers the standard guided score from [Svv17]  $-\nabla_x \log g$  at  $\epsilon = 0$ .

The *safe-guarded guided proposal*  $P^{g+\epsilon}$  is the law of

$$dX_t = b(t, X_t) dt + Q r_\epsilon(t, X_t) dt + q dW_t, \quad X_0 = x_0. \quad (11)$$

Thus, similar to the discrete-time case, the safe-guarded process can be viewed as a mixture of the measure  $P$  and  $P^g$ .

**7.1. IS weight and second moment  $m(\epsilon)$ .** The importance sampling (IS) weight for a  $x \sim P^{g+\epsilon}$ , path initialised at  $x_0$ , is

$$G_\epsilon(x_0, x) = h(T, x_T) \frac{dP}{dP^{g+\epsilon}}(x) = h(T, x_T) \frac{g(0, x_0) + \epsilon}{g(T, x_T) + \epsilon} \cdot \Psi(x).$$

Note that we added  $\epsilon$  as a subscript to  $\Psi$  to highlight this dependence. The IS estimator  $\hat{L} = \frac{1}{N} \sum_{n=1}^N G_\epsilon(x_0, X_T^n)$  with  $X^n \sim P^{g+\epsilon}$  is unbiased for  $h(0, x_0)$  for any  $\epsilon \geq 0$ . Its variance is given by

$$\begin{aligned} m(\epsilon) &= \mathbb{E}_{P^{g+\epsilon}} G_\epsilon(X)^2 = \int G_\epsilon(x_0, x)^2 dP^{g+\epsilon}(x) \\ &= \int G_\epsilon(x_0, x)^2 \frac{dP^{g+\epsilon}}{dP}(x) dP(x) \\ &= \int \frac{dP}{dP^{g+\epsilon}}(x) h(T, x_T)^2 dP(x) \\ &= (g(0, x_0) + \epsilon) \int \frac{h(T, x_T)^2}{g(T, x_T) + \epsilon} \Psi_\epsilon(x) dP(x). \end{aligned} \quad (12)$$

This is to be compared to (6):  $c$  in that formula is exactly as  $g(0, x_0)$  here. The term  $\Psi(x)$  in the integral is an additional path-space correction. In the discrete-time we assumed linear dynamics and then choosing  $\tilde{b} = b$  results in  $\Psi_\epsilon(x) = 1$ .

**7.2. Adaptive choice of  $\epsilon$ .** We minimise  $m(\epsilon)$  over  $\epsilon \geq 0$ . Since the integral in (12) is under  $P$  (not  $P^{g+\epsilon}$ ), we can estimate  $m(\epsilon)$  using  $\{X^n\}_{n=1}^N$  drawn from  $P$ , which are independent of  $\epsilon$ . Weighting by the current normalised particle weights  $\{W^n\}_{n=1}^N$  gives the empirical estimator

$$\hat{m}(\epsilon) = (g(0, x_0) + \epsilon) \sum_{n=1}^N W^n \frac{h(T, X_T^n)^2}{g(T, X_T^n) + \epsilon} \cdot \Psi_\epsilon(X^n),$$

where  $\Psi_\epsilon(X^n)$  is evaluated by discretising (10) along the stored path:

$$\log \Psi_\epsilon(X) \approx - \sum_{\ell=0}^L (b(t_\ell, X_{t_\ell}) - \tilde{b}(t_\ell, X_{t_\ell}))^\top r_\epsilon(t_\ell, X_{t_\ell}) \Delta t.$$

Since both the paths  $X^n$  (sampled under  $P$ ) and the factor  $(b - \tilde{b})^\top \nabla_x \log g \cdot \Delta t$  are independent of  $\epsilon$ , only the factor  $g/(g + \epsilon)$  in  $r_\epsilon$  changes across the grid, making evaluation of  $\hat{m}(\epsilon)$  cheap via the precomputed quantities

$$\delta_\ell^n = (b(t_\ell, X_{t_\ell}^n) - \tilde{b}(t_\ell, X_{t_\ell}^n))^\top r_0(t_\ell, X_{t_\ell}^n) \Delta t, \quad g_\ell^n = g(t_\ell, X_{t_\ell}^n),$$

so that  $\log \Psi_\epsilon(X^n) = -\sum_\ell \delta_\ell^n g_\ell^n / (g_\ell^n + \epsilon)$ . The optimal  $\epsilon^*$  is found either by grid search or, using the analytical derivative **double check**

$$\frac{d\hat{m}}{d\epsilon}(\epsilon) = \sum_{n=1}^{N_{\text{aux}}} W^n \frac{h(T, x_T^n)^2 \Psi_\epsilon(x^n)}{g(T, x_T^n) + \epsilon} \left( 1 - \frac{g(0, x_0) + \epsilon}{g(T, x_T^n) + \epsilon} - (g(0, x_0) + \epsilon) \frac{d \log \Psi_\epsilon}{d\epsilon} \right),$$

by Brent’s method applied to  $d\hat{m}/d\epsilon = 0$ , which is significantly faster in practice.

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## APPENDIX A. ADAPTIVE CHOICE OF $\epsilon$ VIA ESS TARGETING

**This does not work; I guess the desired ESS simply cannot be attained.**

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